in nearly all modern devices that use an integrated circuit. In essence the operating system is just another program, albeit vastly more complicated. The function of the operating system is to organise the rest of the programs used on a computer and manage the hardware resources in relation to those programs.

The operating system is the first boundary of contact between user commands and the hardware that executes them. It allows users to interact and manipulate the hardware to provide desired results. Its important to realise that every click of the mouse and tap of the keyboard is an interaction with the hardware, which we see depicted on the visual display in many appealing styles. The only way an operating system program can interact with the hardware is if they speak the same language, this is accomplished with the use of “drivers”. The concept of the driver was invented so that different operating systems could interface with different types of hardware without any problems. Drivers are nothing but lines of code that can be deployed or removed based from the operating system to alter its interaction with the hardware. These drivers can vary radically from system to system - so we will only discuss the specific ones for Raspberry Pi.

What is Assembly Code?

In any standard computer there is a processor and some memory called RAM (Random Access Memory). The processor is the tool that performs simple functions like math, while the RAM is used to store numbers. The processor is tasked by the programmer to execute a series of instruction and interact with the hardware, this process causes the numbers to be changed in the RAM. These instructions are then translated to readable text for humans thanks to assembly code. Assembly code is the nearest and closest computer language that we can use to instruct the hardware. The reason that it is only an approximate representation of the communication is because of how computers are designed at a fundamental level in a mathematical language.

In usual programming practice, code is written in an assortment of languages such as C, C++, Java and Basic. After the programmer has written code in any of these languages it needs to translated into assembly code. The tool used to accomplish this translation is known as the compiler. The mathematical language of the computer is actually in binary which is very difficult for humans to read. Assembly code on the other hand is more readable but limited in its expressions but cleanly understandable